

ELECTRONIC SALES AND REVENUE ANALYSIS

DATA ANALYTICS PROJECT DOCUMENTATION

Project Type: Data Analytics Project Dataset
Data Period: January 2019 – December 2019

Source: Kaggle

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1. Project Title

Electronic Sales and Revenue Analysis

2. Project Objective

Our main goal with this project was to take raw e-commerce sales data and turn it into insights that actually mean something for a business. We wanted to understand how sales were performing overall and dig into the patterns behind product sales, geographical locations, purchase timing, and customer buying behavior. We used Python and Pandas to clean and explore the data, which helped surface trends and opportunities that could support better business decisions. To make everything easy to explore and present, we brought the results together in an interactive Power BI dashboard.

3. Problem Statement

- E-commerce businesses generate a huge amount of sales data — details on products, orders, prices, purchase locations, and how customers shop — but on its own, this raw data doesn't reveal much.
- It doesn't clearly show which products actually drive the most revenue versus which ones just sell in high volume.
- It's not obvious which locations contribute the most to sales without digging into the numbers.
- There's no easy way to tell when customers are most likely to place orders or which products tend to get bought together.
- Without proper cleaning and analysis, it's hard to make real use of the data for decision-making — which is exactly what this project set out to fix by uncovering the patterns that matter and turning them into insights the business can act on.

4. Dataset Information

Dataset Attribute	Details
Dataset Name	E-Commerce Sales Dataset
Total Records	185,980 rows
Total Columns	11 columns
Data Period	January 2019 to December 2019
Geographical Coverage	9 cities across 8 states in the United States
Data Source	Kaggle
Row Meaning	Each row represents a product line item within an order
Important Columns	Order ID, Product, Quantity Ordered, Price Each, Order Date, Purchase Address, Month, Sales, City, Hour

The dataset is made up of order-level sales records, where each row really represents a single product line item within an order. That means one order can span several rows if a customer bought more than one product, since they'd all share the same Order ID. It includes information on products, quantities, prices, sales figures, order dates, purchase locations, and time-related details — everything needed to clean, explore, and pull meaningful business insights from the data.

5. Tools and Technologies Used

Tool / Technology	Purpose
Python	Used as the primary programming language for data cleaning, analysis, and exploratory data analysis.
Pandas	Used for data manipulation, cleaning, transformation, and analysis of the dataset.
Jupyter Notebook	Used as the development environment to carry out and document the data analysis process.
Power BI	Used to build an interactive dashboard for visualizing key sales insights and business performance.

6. Data Cleaning

Before we could explore the data, we cleaned and validated it using Python and Pandas. The raw dataset started out with 185,980 rows and 11 columns.

6.1 Missing Value Handling

We found missing values scattered across a few key fields — 20 records were missing data in Product, Quantity Ordered, Price Each, Sales, or City. Since these records were incomplete, we removed them so the rest of the analysis would be based on complete, reliable observations.

6.2 Duplicate Record Handling

An initial duplicate check turned up 30 exact duplicate records, which we removed. We kept checking for duplicates after later transformations too. A few more rows started to look like duplicates once certain identifying fields had been dropped, but we didn't delete these outright — a single order can legitimately contain several different product line items, so what looks like a duplicate isn't always one.

6.3 Date and Time Processing

We converted the Order Date field into a proper datetime format and then split it into separate Order_Date and Order_Time columns, which made time-based analysis much easier to work with.

6.4 Data Type Correction

Quantity Ordered was stored as a floating-point number, which didn't really make sense since order quantities are always whole units. We converted it to an integer to fix that.

6.5 Unnecessary Column Removal

Once we'd pulled what we needed from them, we dropped the Month column, the original Order Date column, and an unnamed index column that had come in with the file. We also renamed Hour to Order Hour so it would be clearer what it represented.

6.6 Data Validation

We checked Quantity Ordered, Price Each, and Sales for anything invalid or non-positive, and didn't find any issues. We also cross-checked Sales against Quantity Ordered $\times$ Price Each — the largest difference we found was about 5.68×10^{-14} , which is just floating-point rounding noise, confirming the Sales values were consistent.

6.7 Geographical Data Preparation

We extracted the state from the Purchase Address field and mapped the state abbreviations to full state names, which set us up to do proper geographical analysis later on.

6.8 Outlier Analysis

We ran an IQR-based outlier check on Sales, which flagged 37,015 records — about 19.9% of the dataset — as potential outliers. We chose to keep them, though, since the dataset naturally mixes low-priced accessories with high-value electronics, so a lot of those "high" sales values are perfectly legitimate business observations. Removing them would have distorted the revenue picture rather than cleaned it up.

Once the data was cleaned and validated, we moved on to exploratory data analysis and building the Power BI dashboard.

7. Data Analysis / Exploratory Data Analysis

With the data cleaned, we used Pandas, Matplotlib, and Seaborn to explore product performance, sales trends, geographical distribution, order timing, pricing and quantity patterns, and which products tend to sell together.

7.1 Product Performance Analysis

Top products by revenue:

Rank	Product	Revenue
1	MacBook Pro Laptop	\$8,037,600
2	iPhone	\$4,790,800
3	ThinkPad Laptop	\$4,128,959
4	Google Phone	\$3,316,800
5	27in 4K Gaming Monitor	\$2,437,048

Top products by quantity sold:

	Product	Units Sold
1	AAA Batteries (4-pack)	31,004
2	AA Batteries (4-pack)	27,648
3	USB-C Charging Cable	23,979
4	Lightning Charging Cable	23,224
5	Wired Headphones	20,558

What's interesting is that the top five products by revenue and the top five by quantity don't overlap at all. That tells us high-ticket electronics are the real revenue drivers, while cheaper accessories are what's actually moving in the highest volume.

7.2 Sales Trend Analysis

We looked at how sales activity moved across 2019 using daily observations. Overall, things picked up in the second half of the year, with a noticeable jump in the fourth quarter and especially in December.

7.3 Geographical Sales Analysis

Top cities by sales:

Rank	City	Sales
1	San Francisco	\$8,259,164
2	Los Angeles	\$5,451,959
3	New York City	\$4,664,732
4	Boston	\$3,661,549
5	Atlanta	\$2,794,295

State-level sales distribution: California is clearly the strongest market here by a wide margin.

State	Sales	Share
California	\$13,711,123	36.9%
New York	\$4,664,732	12.6%
Texas	\$4,585,035	12.3%
Massachusetts	\$3,661,549	9.9%
Georgia	\$2,794,295	7.5%
Washington	\$2,747,446	7.4%
Oregon	\$1,870,464	5.0%
Maine	\$453,178	1.2%

7.4 Time-of-Day Analysis

We also broke sales down by hour of the day. 7 PM (19:00) came out as the peak, with roughly \$2.41 million in cumulative sales. Activity was minimal in the early morning, started climbing from around 7 AM, and stayed strong through most of the 11 AM–9 PM window.

7.5 Quantity and Price Analysis

Most orders were for just a single item — quantity-of-one orders accounted for about \$34.1 million of the roughly \$34.5 million total sales figure. Price Each and Sales turned out to be almost perfectly correlated ($r \approx 0.999$), which tracks given how many orders are for a single unit. Quantity, on the other hand, showed a weak negative correlation with both Price Each ($r \approx -0.15$) and Sales ($r \approx -0.14$), and Order Hour showed almost no correlation with any of the numeric fields.

7.6 Market Basket Analysis

We looked at which product combinations kept showing up within the same Order ID to spot items customers tend to buy together. The strongest pairings we found were:

Product Pair	Pair Count
Lightning Charging Cable + iPhone	1,008
Google Phone + USB-C Charging Cable	994
Wired Headphones + iPhone	462
Google Phone + Wired Headphones	422
Apple AirPods Headphones + iPhone	372
USB-C Charging Cable + Vairebadd Phone	368
Bose + Google Phone	228

These pairings point to some clear opportunities for product bundling and cross-selling.

7.7 Average Order Value

The average order value (AOV) came out to about \$193.37. That's on the higher side, mostly because of how many high-ticket electronics show up in the dataset.

8. Key Insights and Findings

8.1 High-Value Products Drive Revenue

Laptops, smartphones, and monitors are what really drive revenue, even though they don't sell in huge quantities. Batteries, cables, and headphones are the opposite story — cheap, but sold constantly, which is what generates the highest unit volume.

8.2 California Is the Strongest Market

California alone accounts for roughly 36.9% of total revenue, making it by far the leading state in the dataset. No other state comes close, which makes it the clearest priority market to focus on.

8.3 Evening Hours Show Strong Sales Activity

Sales activity is particularly strong through the afternoon and evening, with 7 PM standing out as the peak hour. Activity is minimal overnight and builds steadily from around 7 AM onward.

8.4 Accessories Have High Sales Volume

Low-priced accessories like batteries and charging cables sell in huge quantities compared to everything else in the dataset. Keeping them consistently in stock matters just as much as managing the higher-priced electronics.

8.5 Strong Product Pairing Opportunities

Phone-and-accessory combinations, like a phone bought alongside a charging cable or headphones, show up together often. That's a solid opportunity for bundles and cross-selling promotions built around these natural pairings.

8.6 Revenue Is Influenced by High-Value Electronics

The \$193.37 average order value is pulled up by high-priced electronics rather than reflecting a typical purchase. Most individual line items are priced well below that figure.

8.7 Business Implications

Taken together, these findings support stocking more accessories, building out product bundles, and running evening promotions. They also point toward paying closer attention to the California market and keeping high-value electronics reliably in stock.

9. Power BI Dashboard

To bring everything together, we built an interactive Power BI dashboard that presents the major findings from the analysis in a clear, visual way. Rather than just reading about the results, the dashboard lets you explore them

directly — by product, location, time, and product-pair views.

Dashboard Title: Electronics Sales & Revenue Analysis

9.1 Dashboard Features

- Total Sales – the overall sales value
- Total Orders – the total number of orders placed
- Total Quantity – the total number of units sold
- Average Order Value – the average value per order
- Average Selling Price – the average price per unit sold
- MoM Growth – a month-over-month growth view

- Sales by City – compares sales performance across cities
- Sales by Product – shows product-level sales performance
- Sales by Hour – shows sales activity across ordering hours
- Product Pair Analysis – shows frequently occurring product combinations
- Date/Month Analysis – supports time-based sales exploration

9.2 Dashboard Purpose

The dashboard takes the analytical results and turns them into an interactive, business-friendly view. It makes it easy to quickly see overall performance, spot high-performing products, identify strong markets, understand key sales hours, track growth patterns, and explore product combinations — all in one place.

9.3 Dashboard Screenshot

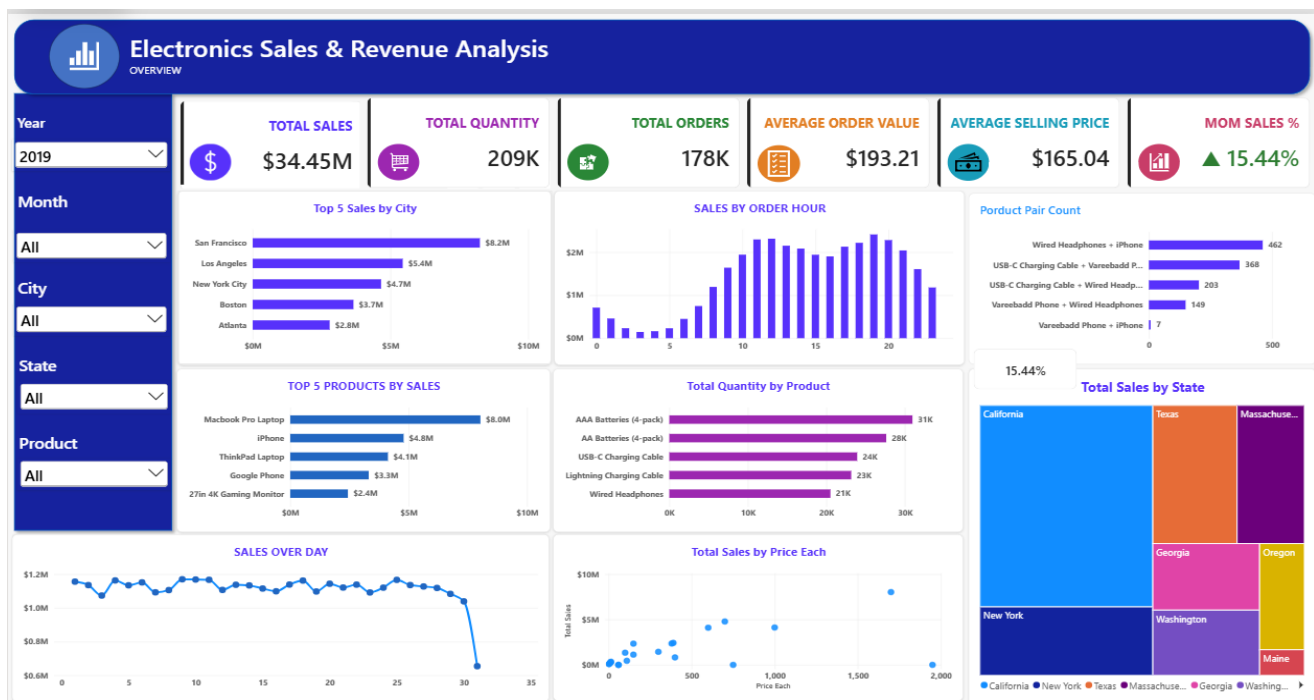


Figure 1: Electronics Sales & Revenue Analysis – Power BI Dashboard

10. Conclusion

In short, this project cleaned and analyzed 185,980 order line items from 2019 using Python, Pandas, and Power BI to uncover what's really driving sales. High-value electronics turned out to be the main revenue drivers, low-priced accessories dominate in volume, California is by far the strongest market, and evening hours see the most sales activity. The resulting Power BI dashboard brings all of these findings together into one easy-to-explore view that can support better inventory, marketing, and bundling decisions.

11. Future Improvements and Limitations

11.1 Limitations

- **Duplicate Handling:** we checked for duplicates at a few different points in the workflow rather than all in one pass right after loading the data — a more consistent approach applied immediately would be cleaner.
- **Sales Trend Granularity:** one of our visualizations was labeled as monthly even though it was actually built from daily observations. That trend should be explicitly aggregated by month if it's going to be presented as a monthly view.
- **Descriptive Analysis:** this project focuses on descriptive and exploratory analysis and doesn't include any predictive modeling.
- **Market Basket Analysis:** products showing up together in the same order tells us they're associated, not that buying one caused the purchase of the other.
- **Geographical Scope:** the dataset only covers U.S. sales, so these findings may not generalize to other markets.

11.2 Future Improvements

- Apply one consistent duplicate-detection approach at the earliest appropriate stage
- Add monthly, quarterly, and yearly comparative analysis
- Build out sales forecasting models for future demand estimation
- Add customer segmentation once customer-level information is available
- Use more advanced association-rule techniques for product recommendation and bundling
- Bring in additional datasets such as customer, marketing, discount, and inventory data
- Extend the project with predictive machine learning techniques
- Enhance the Power BI dashboard with more business-focused views and analytical capabilities

12. Project Developed By

Details	Information
Project Title	Electronic Sales and Revenue Analysis
Project Type	Data Analytics Project
Team Name	Data Pioneers
Team Members	Shiva Prasad, Paramesh, Basheer Ahmed
Course / Program	Data Science
Institution	TEKS Academy
Tools Used	Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook, Power BI

13. References and Dataset Source

13.1 Dataset Source

We sourced the E-Commerce Sales Dataset used for this project from Kaggle.

13.2 Project Resources

- Sales_Data.csv – the source sales dataset used for the project
- Python/Pandas notebook – used for data cleaning, transformation, validation, and exploratory analysis
- Power BI report – used to build the interactive sales and revenue dashboard

13.3 Technologies Referenced

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook
- Microsoft Power BI

END OF PROJECT DOCUMENTATION